PRESENTATION SCRIPT

Smart Farm

What to say on each slide of the 17-slide deck, with timing, cues and answers to likely questions.



Home screen on the 480 × 320 touchscreen

17
slides

~14 min
speaking time, plus questions

10
practice questions with answers

Before you start

- **Speak to the people, not the slide.** The slides carry the details; your job is to tell the story.
- **Lines in *[pink brackets]*** are actions, not words to read out.
- **Share the slides** if two or three people present: for example slides 1–6, 7–12 and 13–17.
- **Keep the real device running** next to you. Showing the touchscreen live after slide 12 or during questions is the strongest part of the talk.
- **Fill in the blanks** marked ____ (your names, the parts cost) before the day.

#	Slide	Time
1	Smart Farm (title)	0:45
2	Why a smart farm	1:00
3	System architecture	1:00
4	Hardware specifications	0:45
5	Wiring and pin map	0:45
6	Software and code structure	0:45
7	How it works	1:00
8	Pump 1: automatic watering	1:00
9	Rain tank alert and pump 2	1:00
10	Daily advice from the forecast	1:15
11	Farm AI with Gemini	1:00
12	Tablet-style user interface	0:45
13	Code highlights	0:45
14	Test results	1:00
15	Who benefits	0:45
16	Conclusion and future work	0:45
17	References and thank you	0:15
	Total	**about 14:30**

1 Smart Farm

~0:45



Good morning, everyone. We are ____ and ____, and today we'd like to show you **Smart Farm**.

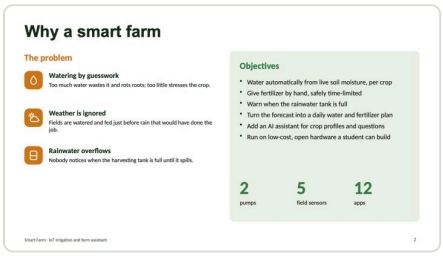
Smart Farm is a small, low-cost system that looks after a crop by itself. It measures the soil, the air and a rainwater tank. It waters the crop when the crop actually needs water, it helps us give fertilizer at the right time, and it reads the weather forecast to give us advice for the day.

[Point to the screenshot] What you see here is the real home screen of our device, running on a 3.5-inch touchscreen.

---

2 Why a smart farm

~1:00



Let's start with the problem. On most small farms, watering happens on a routine: every morning, whether the crop needs it or not. Too much water is wasted and can rot the roots; too little stresses the plant.

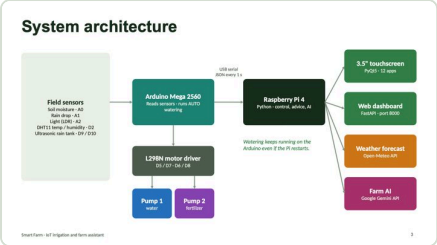
Second, the weather is ignored. We often water or put fertilizer just before it rains, and the rain washes the fertilizer away. Third, rainwater tanks overflow without anyone noticing.

So our objectives were simple: water automatically from the real soil moisture, for each crop; give fertilizer safely; warn when the tank is full; turn the forecast into a daily plan; add an AI assistant; and keep it cheap enough for a student to build.

> The finished system: 2 pumps, 5 field sensors and 12 apps on the screen.

3 System architecture

~1:00



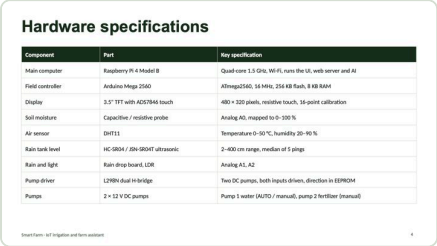
[Follow the arrows from left to right] On the left are the field sensors: soil moisture, rain, light, temperature and humidity, and an ultrasonic sensor on the rainwater tank. They connect to an **Arduino Mega**, our field controller, which also drives the two pumps through a motor driver.

Every second, the Arduino sends all its readings to a **Raspberry Pi** over USB. The Pi runs everything people see: the touchscreen, a web dashboard for any phone, the weather forecast and the AI.

> Design choice: automatic watering runs on the Arduino itself, so the crop still gets water even if the Pi restarts.

4 Hardware specifications

~0:45

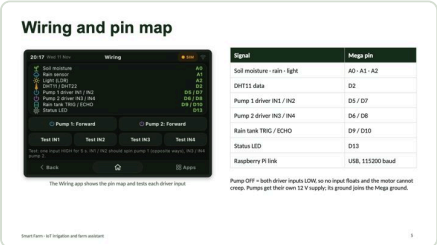


Here are the parts. A Raspberry Pi 4 is the main computer, an Arduino Mega 2560 is the field controller, and a 3.5-inch touchscreen is the display. The sensors are common, inexpensive modules, and the two 12-volt pumps run through an L298N motor driver.

Everything here is easy to buy and easy to replace. The parts cost us about ____ in total.

5 Wiring and pin map

~0:45



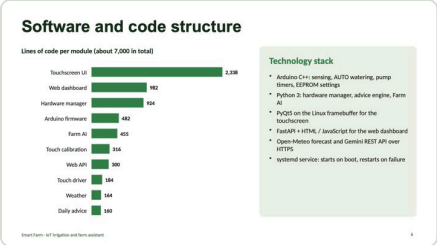
This is the pin map, and the same map is built into the device in its Wiring app.

One lesson from testing: our pump kept turning slowly even when it was switched off. One input of the motor driver was left unconnected and picked up electrical noise. Now the Arduino drives **both** inputs of each pump, and when a pump is off, both are held low, so it stays completely off.

The Wiring app can also test each input one by one. That is how we found two crossed wires and fixed them in software.

6 Software and code structure

~0:45

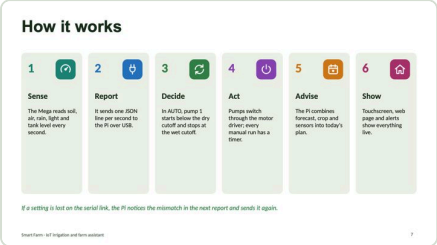


The project is about **7,000 lines of code**. The biggest part is the touchscreen interface: a resistive screen only understands taps, so every control had to be a large button or a plus-and-minus stepper.

The Arduino is programmed in C++, the Pi runs Python, the web dashboard uses FastAPI, and the whole system starts by itself when the Pi is switched on.

7 How it works

~1:00



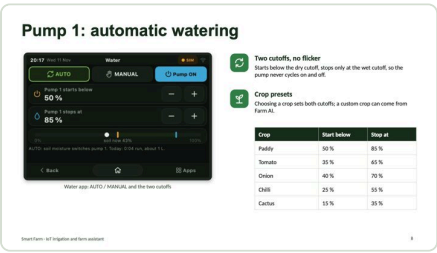
This cycle runs every second. **One, sense:** the Arduino reads every sensor. **Two, report:** it sends one line of data to the Pi. **Three, decide:** in AUTO mode, if the soil is drier than the lower limit, pump 1 starts. **Four, act:** the pump runs, and every manual run has a timer, so nothing is left on by mistake.

Five, advise: the Pi combines the forecast, the crop and the sensors into a plan for the day. **Six, show:** everything appears live on the screen and the web page.

And if a setting is ever lost on the way to the Arduino, the Pi notices within seconds and sends it again.

8 Pump 1: automatic watering

~1:00



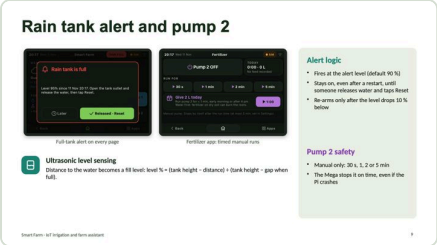
Pump 1 waters the field, and it uses **two limits, not one**. For paddy, it starts when the soil drops below 50 percent and stops only when it reaches 85 percent.

Why two? With a single limit, the pump would keep switching on and off around that one value. With a gap between the limits, it runs in one steady cycle and then rests.

[Point to the table] When you choose a crop, both limits are set automatically, and you can still fine-tune them on the screen.

9 Rain tank alert and pump 2

~1:00



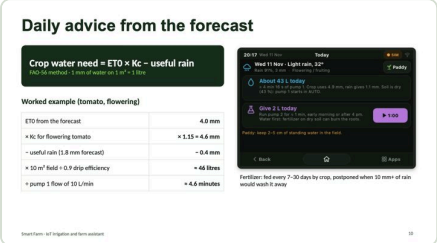
The rainwater tank is measured with an **ultrasonic sensor**. It sends a sound pulse down to the water and times the echo, the same way a bat judges distance.

When the tank reaches 90 percent, a red alert appears on every page of the touchscreen and on the web page. It stays until someone really releases the water and presses Reset, even if the device restarts.

Pump 2 is for liquid fertilizer, and it is manual on purpose. You choose 30 seconds, 1, 2 or 5 minutes, and the Arduino switches it off on time, even if the Pi crashes.

10 Daily advice from the forecast

~1:15



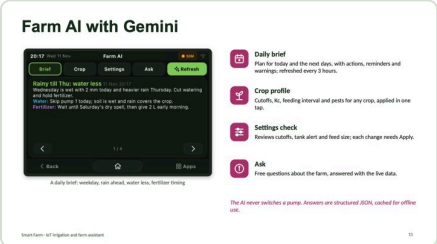
This is where the weather comes in. Every day the system downloads the forecast, which includes a number called **ET0**: how much water a reference crop loses today through evaporation.

We multiply it by a crop factor, **Kc**, and subtract the rain that is expected. *[Point to the table]* For a flowering tomato: 4 millimetres times 1.15 is 4.6; the forecast rain gives back about 0.4. Over a 10-square-metre field that is about 46 litres, or four and a half minutes of pumping.

For fertilizer, it remembers when the crop was last fed, and if heavy rain is coming it tells you to wait, because the rain would wash it away. The method comes from the FAO, the United Nations food and agriculture organisation.

11 Farm AI with Gemini

~1:00



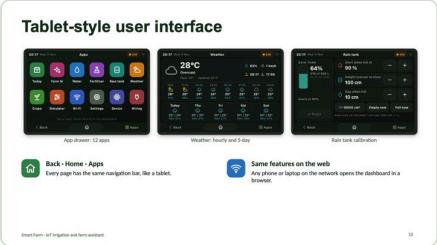
On top of these rules, we added an AI assistant using Google's **Gemini**. Every few hours the Pi sends it the date, the forecast, the sensor readings and the crop, and it writes a short brief, like this one: "*Rainy till Thursday: water less.*"

It can also build a profile for any crop you type in, check whether our settings make sense, and answer questions about the farm.

Safety rule: the AI only suggests. It never switches a pump; a person always has to press Apply.

12 Tablet-style user interface

~0:45



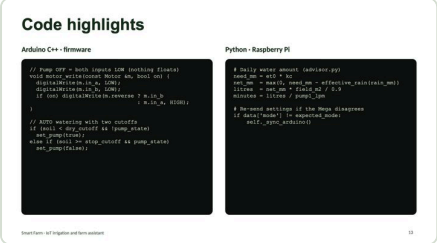
The interface is designed like a tablet: an app drawer with 12 apps, and the same Back, Home and Apps bar on every page. Here is the weather app, and here is the rain tank setup.

Everything on the touchscreen is also on a web page, so a farmer can check the farm from their phone.

[Good moment for a 30-second live demo: open the Apps drawer, then Today, then go back Home]

13 Code highlights

~0:45



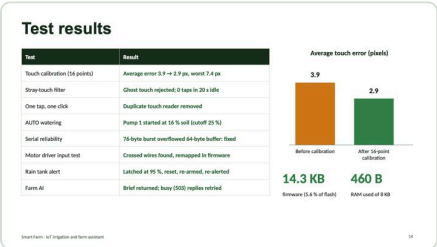
Just two small pieces of code, because they carry most of the logic.

[Left panel] On the Arduino, switching a pump always sets both driver inputs low first; that is the fix for the creeping pump. Below it is the two-limit rule: start below the dry limit, stop at the wet limit.

[Right panel] On the Pi, today's water: crop need minus useful rain, scaled by the field size and divided by the pump's flow.

14 Test results

~1:00



We tested everything on the real hardware. Touch accuracy improved from **3.9 to 2.9 pixels** after calibration. Automatic watering started the pump at 16 percent soil moisture, below the 25 percent limit we had set.

We also found a communication fault: settings were lost because we sent more data than the Arduino could hold at once. We fixed it and added a check that resends anything that goes missing.

The tank alert, the motor test and the AI were all checked on the device, and the whole Arduino program fits in just 14 kilobytes.

15 Who benefits

~0:45

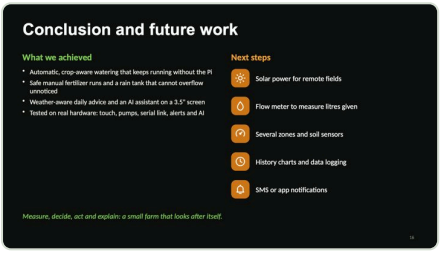


Who is this for? **Small farmers** save water, fertilizer and time. **Students and teachers** get a complete STEM project: sensors, electronics, code and AI in one. **Community gardens** can share one system.

It supports **water conservation** by using rainwater first. **Agricultural extension workers** can share the daily plans. And **the crop itself** gets steady moisture and well-timed feeding.

16 Conclusion and future work

~0:45



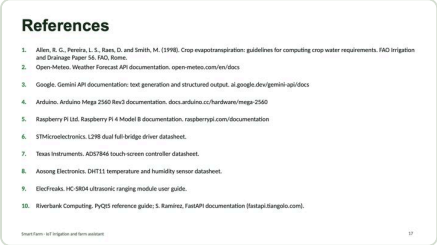
To sum up: we built a system that waters automatically for each crop and keeps working without the Pi, gives fertilizer safely, never lets the rain tank overflow unnoticed, and explains its advice using the weather and AI.

Next, we'd like to add solar power, a flow meter to measure exactly how much water is given, more zones and sensors, history charts, and SMS alerts.

> Closing line: "Measure, decide, act and explain: a small farm that looks after itself."

17 References and thank you

~0:15



Our main sources are listed here: the FAO method for crop water needs, and the documentation for the hardware and services we used.

Thank you for listening. We're happy to answer your questions, and we can show the system working live.

Likely questions and answers

Short, honest answers. If you don't know something, say so and offer to show the device.

What happens if the internet goes down?

Watering keeps working: it runs on the Arduino from the soil sensor and needs no internet. The forecast and AI show their last saved results until the connection returns.

What if the Raspberry Pi crashes?

The Arduino keeps AUTO watering going on its own, and it stops pump 2 when its timer ends. When the Pi comes back, it re-sends all settings automatically.

Why use two controllers instead of one?

The Arduino is simple and reliable for real-time sensing and switching pumps. The Pi is a full computer for the screen, the web page, the forecast and the AI. Splitting them means a software problem on the Pi can't stop the watering.

How does the ultrasonic sensor measure the water?

It sends a sound pulse and times the echo. Distance = time × speed of sound ÷ 2. The system turns that distance into a fill percentage using the tank's height.

How accurate is the soil moisture reading?

It's a relative reading, not a lab value. That's why each crop has a start and a stop limit you can adjust: you watch the reading when the soil is dry and when it's well watered, and set the limits between them.

What happens if the temperature sensor stops working?

It retries and keeps the last good reading for two minutes. After that, the screen shows the weather station's value, clearly labelled "weather", so you always know where the number comes from.

Can the AI give wrong advice?

Yes, it can. That's why it only suggests and never controls a pump, and why the rule-based daily advice works without it. Any fertilizer or chemical dose should be checked with an agriculture officer.

Why does the fertilizer pump have no automatic mode?

Too much fertilizer can burn the roots, so a person should decide. The system tells you when it's due and how long to run the pump, but you press the button.

How much did it cost?

About ____ for all the parts. Most of it is the Raspberry Pi and the screen; the sensors are inexpensive modules.

Could this work on a bigger farm?

Yes, with more zones: more soil sensors and pumps, and a flow meter to measure water exactly. Those are on our list of next steps, along with solar power for fields without electricity.